

Costs & revenue

Objectives of firms

Survival: When firms are started, their initial objective may be just to survive in what may be a very competitive market. A firm may be content to just cover its costs until it can become better known. During difficult times when demand is falling, even large firms may have survival as their key objective. They will try to stay in the market in the hope that conditions will improve.

Growth: Some firms may pursue the objective of growth. Increasing the size of the firm may bring a number of advantages. High and expanding sales tend to enable firms to take advantage of a number of internal economies of scale and so, for instance, to raise finance more easily and to buy raw materials at a discounted rate. Those who run firms, the managers, directors and chief executives, may have the growth of the firm as their key objective because their pay and status may be more closely linked to the size of the firm they run. Those in charge of large firms are usually paid more than those running smaller firms and also tend to be held in higher esteem. They may also have greater job security as the larger the firm they run, the more difficult it will be for any other firm to take it over and replace them with their own managerial team. If growth is achieved by merging with other firms, competition will be reduced and the firm will gain a larger market share.

Social welfare: State-owned enterprises may be given by the government the objective of improving social welfare. They may, for instance, charge a relatively low price for their products to ensure they are affordable to even the poor. They are more likely than private sector firms to base their production decisions on social costs and benefits. In recent years, some private sector firms have also been showing a greater concern about the environmental and social effects of their actions. A number of firms have sought to clean up their production processes and ensure that they source their raw materials from firms that do not employ child labour.

Profit satisficing: In some cases, firms may engage in what economists call profit satisficing. This involves making enough dividends to keep shareholders happy while pursuing other objectives. For example, a firm may be prepared to sacrifice some

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profit, at least in the short run, in order to improve staff facilities or to get their raw materials from more sustainable sources.

Profit maximization: Traditional theory, however, suggests that firms seek to maximize profits. This means that they try to earn the largest profit possible over a period of time. Whilst some of the other objectives may appear to conflict with profit maximization, pursuit of them may actually increase profits in the longer term. For instance, growth may involve reducing the number of competitors. This would reduce the price elasticity of demand for its product and so its ability to raise price and revenue. In addition, increasing the scale of operation may reduce average costs. Both of these outcomes would increase profits.

Seeking to increase social welfare may also increase profits in the long run. For example, pursuing environment friendly policies and being socially responsible may raise costs of production, but it may also increase demand and revenue, as consumers are becoming increasingly concerned that firms should act in an ethical manner. Treating workers better may help retain workers which would reduce the costs of recruiting and training workers.

When profit maximization is achieved

Profit is made when the revenue earned by a firm is greater than the costs incurred by it. Profit maximization is an objective pursued by most private sector firms. Total profit is the positive difference between total revenue and total cost. Profit per unit (sometimes referred to as the profit margin) is the positive difference between average revenue (revenue per unit) and average total cost (unit cost). Profit is maximized when the positive gap between revenue and cost is greatest. Table 22.6 shows that profit would be maximized at 40 units of output.

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Output	Total revenue (\$)	Total cost (\$)	Total profit (\$)
10	200	220	-20
20	380	380	0
30	500	480	20
40	600	540	60
50	660	620	40
60	700	710	-10

Table 22.6 The relationship between total revenue, total cost and total profit

Besides calculating total profit, the profit per unit can be found by deducting average cost from average revenue, as shown in Table 22.7. In this case, profit is maximized at three units.

Output	Average revenue (\$)	Average cost (\$)	Total profit (\$)
1	15	15	0
2	14	12	2
3	12	9	3
4	9	8	1
5	5	10	-5

Table 22.7: The relationship between average revenue, average cost and total profit

From the information in Table 22.7, it is possible to calculate total revenue (by multiplying average revenue with output), total cost (by multiplying average cost with output) and total profit (by multiplying profit per unit with output). These figures are shown in Table 22.8.

Output	Total revenue (\$)	Total cost (\$)	Total profit (\$)
1	15	15	0
2	28	24	4
3	36	27	9
4	36	32	4
5	25	50	-25

Table 22.8: Total revenue, total cost and total profit

If information is given on output, costs and revenue, it is possible to work out revenue as shown in Table 22.9. This is because it is known that

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$$\text{Profit} = \text{revenue} - \text{cost}$$

So, revenue = profit + cost. Similarly, from information on output, revenue and profit, cost can be calculated.

$$\text{Cost} = \text{revenue} - \text{profit}$$

Output	Total profit (\$)	Total revenue (\$)	Total cost (\$)
10	0	400	400
20	100	800	700
30	300	1200	900
40	400	1600	1200
50	300	2000	1700

Table 22.9: The relationship between total profit, total revenue and total cost

Effects of changes in profits

Profits provide an incentive for entrepreneurs to undertake production. An increase in profit will encourage more firms to enter a competitive market. It will also provide firms with more finance to update their capital equipment and expand their business.

A profitable firm will also find it easier to obtain external finance. Shareholders are more likely to want to buy shares in profitable firms and banks are usually willing to give them loans. These firms may also find it easier to recruit top managers and directors, attracted by their success.

The effect of a fall in profit may vary with time. At first it may have little impact on the behavior of firms, if they think that it will only be short lived. After a while, if profits remain low, or fall further, some firms will cut back on production and others will cease production.

Ways of increasing profit

The two fundamental ways of increasing profit are to

- reduce costs of production, and
- raise revenue.

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There are a number of ways of reducing costs of production. One is by reducing any wastages and inefficiency. Another is by increasing the productivity of factors of production. In the short run the second strategy may actually raise costs, but in the long run it may lower average costs and raise revenue by improving quality. For instance, a firm may spend more on training workers and may replace existing equipment with a more technologically advanced version. In the longer run, these measures should increase output per worker and per machine and therefore reduce average cost. A third way is by increasing the size of the firm through merger or takeover. A larger firm may be able to take advantage of economies of scale. The firm will be likely to have higher total revenue. It may also have a higher total profit and profit per unit, due to its greater market share. Firms with considerable market power often have inelastic demand for their products. When demand is inelastic, a firm can increase its revenue by raising price. In contrast if demand is elastic, the revenue may be raised by cutting down price.

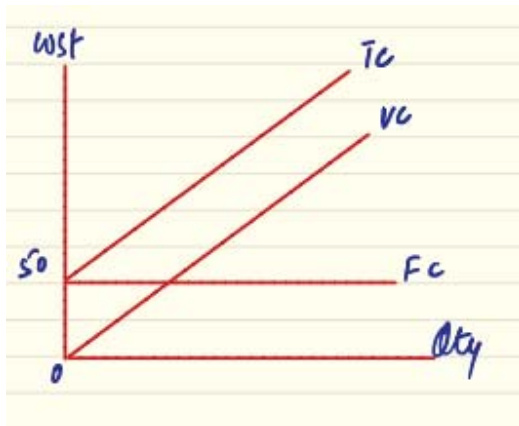
Besides trying to raise revenue by changing price, firms may seek to increase demand for the products. There is a variety of ways in which this can be done. They may seek to improve the quality of their products, diversify and be more responsive to changes in consumer demand by improving their market research. Another method that firms may employ is advertising. A successful advertising campaign is one which increases revenue by more than the cost and hence raises profit.

COSTS:

FIXED COSTS: These are the costs associated with the fixed factors of production and do not vary with the output. For example, rent, rates, insurance etc. whether the firm operates at full capacity or less than full capacity, these cost have to be paid.

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Let us look at a business such as the Suzuki Motor Company. The company has big machines, thousands of parts, huge buildings and a multitude of other pieces of plant and equipment that have already been bought. Suzuki has to take account of the wear and tear of this equipment no matter how many cars the company produces. The payments on the loans taken out to buy the equipment and the rent of the land have to be paid regardless of output. All these costs are unaffected by variations in the quantity of output, i.e. they are mainly the overhead costs or fixed costs.

VARIABLE COSTS: These are the costs associated with the variable factors of production i.e. they vary with the production. They are sometimes referred to as operating costs e.g. wages of labor, costs of raw materials, transport cost and other utilities. When output is zero there is no variable cost incurred but as the producer increases output in the short run the variable costs start increasing with the output.

TOTAL COST: It is simply the sum of the fixed and the variable costs. The shape of the total cost curve is the same as the variable cost curve with a vertical distance equal to the fixed cost.

SHORT RUN AVERAGE COST CURVES:

The average cost concept is simply the one in which we are measuring the cost per unit of output.

- Average total cost or Average cost = total cost / output
- Average variable cost = total variable cost / output

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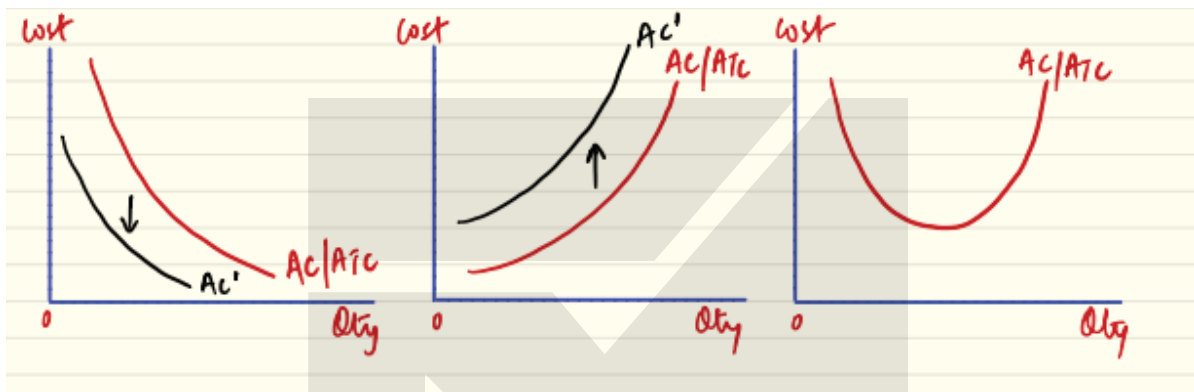
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- Average fixed cost = fixed cost / output

Economies and diseconomies of scale

Economies of scale: Falling of average costs are an important benefit to a firm.

We say that the firm experiences economies of scale, if it makes cost saving from increasing the scale of production that is by raising output.



Diseconomies of scale: If the average cost of production rise as more is produced then the firm is experiencing diseconomies of scale, because the firm is producing too much and has become inefficient.

Optimum size: Optimum size for a firm is where it can reduce average cost to their lowest point in the long run. Here it can benefit fully from economies of scale with the average cost low, profits will be bigger.

Internal economies of scale: Reduction in long run average costs as the scale of production and output of the firm increases. When a firm expands the scale of production it has a choice to become more efficient and lower its average costs. This is because it gives the management or owners a chance to reorganize the way the firm is run and financed. Such decisions are taken within the firm and so the advantages or economies they bring are known as internal economies.

Advantages of large scale production / types of internal economies of scale:

1. **Financial economies:** Large firms are able to raise money from the general public by selling share through the stock exchange. Large firms represent

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low risk borrowers, financial institution may not charge them so much for giving them a loan i.e. lower interest rate.

2. **Purchasing & Marketing economies:** It arises because a large manufacturer can buy in bulk and can therefore get lower prices than the small manufacturer. The cost of selling is also lower, because advertising costs can be spread over a large number of units' sold and specialist sales men can be employed.
3. **Risk bearing economies:** It arises when sells in more markets and has a wider product range than a smaller company. Large enterprises often produce a whole variety of goods and services, if the demand for one falls they still have other they can make and sell. This is known as diversification which means producing a diversity or whole variety of products.
4. **Technical economies:** Large firms can afford to employ specialist workers and machines. They can divide up the production process into specialized tasks so that production becomes faster as each worker becomes an expert in a particular job. Large firms can also afford to research and develop new faster methods of production and new products.

External economies of scale: Those advantages in the form of lower average cost which a firm gains from the growth of industry. All the firms in an industry tend to locate together in one particular place.

1. **Skilled labor:** When firms involved in the same type of activities locate near each other, they all employ and train local people in the work skills they need. A large skilled labor force emerges which can benefit other firms who move into the area.
2. **Ancillary firms:** In areas where similar firms locate other firms may join them to cater for some of their needs. For example, smaller car producing firms have set up near the large car factories that dominate the midlands in the United Kingdom.
3. **Co-operation:** Where firms locate together to produce one particular good or service they tend to help each other even though they are also competing with each other to sell their products.

The disadvantages of large scale production

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Diseconomies of scale

Internal diseconomies of scale:

1. The main cause of these diseconomies appears to be increasing problems faced by **management** as firms grows larger. Larger firms are usually divided into specialist departments. There are departments responsible for sales, production, stock control, accounting, buying personal and so on. It is not easy to organize and supervise the activities of several departments and to make sure that these departments are working together as a team. Weakness in organization and control can be a cause of inefficiency and hence increasing costs.
2. A major challenge facing the managers of larger firms is to find ways of obtaining the willing co-operation of workers. Individual workers in a labor force of several thousand may find it difficult to identify themselves with the firms, they do not feel that they are part of the firm, where management is weak and inefficient workers may come to believe that as far as the firm is concerned they are little more than **numbers on the wage sheet**. Such a state of affair is not likely to lead to efficient and low cost of production.

External diseconomies

1. It will affect all the firms in an industry. They can occur when the growth in a particular area reaches a stage when **increasing congestion raises transport cost**.
2. The growing concentration of an industry will lead to an increasing demand for industrial sites in the area where it is located. This in turn will tend to increase the **price of land, rents, industrial and commercial buildings**.

Small firms: Some of small firms today could grow to become the large firms of tomorrow, but increasingly small firms face tough competition from larger firms. Before we look at how small firms manage to survive, it is important to consider just exactly what is meant by a small firm

- A small firm has a small share of a market

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- It is managed by its owners in a very personalized way
- It is independent, meaning that the small firm is not part of a large company

Large firms can enjoy special advantages like bulk buying, receiving loans at lower interest rates and employing specialist staff with lower average costs than small firms; larger firms can lower their prices to attract consumers. With such fierce competition why is that small firms can still survive and why does there appear to be a growth in their numbers.

1. **The size of the market may be small:** When there is only a small number of consumers willing to buy a product there is no point of growing the firm too large.
2. **The market is local:** Where only one small firm is able to supply a good or service to the people of a particular area we call it a local monopoly
3. **A wide variety of goods and services are wanted:** Large firms that mass produces goods and services cannot afford to keep changing colors and designs. Smaller firms however can often cater for a variety of tastes, tailors can make suits to measure, and carpenters can make furniture order.
4. **Expensive jewelry, luxury cars and yachts** will be produced by small firms for a small number of people
5. People like **personal service**
6. A large firm requires **components parts:** Many small firms can survive by producing parts for large manufacturers
7. Small firms can **co-operate:** Co-operation between small firms can lead them to set up jointly owned enterprises which allow them to enjoy many of the economies of scale that large firms have.
8. **Government assistance**

Growth of firms

Motives behind growth:

- Desire to achieve economies of scale
- Desire to obtain greater market share and market power
- Desire to achieve greater security by extending the range of products and markets.

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Methods: There are two methods of achieving growth

1. **Internal growth or organic growth**, where firms increase its size by making more of its existing products or by extending the range of its products. This type of growth takes place under the same management and structure.
2. **External growth:** It is a common method nowadays. It is achieved by a merger or takeover. A merger is possible when two firms agree to combine to create new firms while a takeover means that a firm asset and the whole firms have been taken over by the firm carrying out the takeover. Growth by merger is referred as **integration** which comprises of vertical, horizontal, or conglomerate.
 - **Vertical integration:** It is when a merger takes place between firms engaged in different production stages. It is vertical in the sense that the merger is a movement up or down the production process which runs from extraction to distribution for e.g. an oil refinery takes over an oil extracting company.
 - **Backward vertical integration:** Vertical integration backwards is when a merger is towards the sources of supplies for e.g. the example given above. It is carried out so that a firm has a greater control over the quality and the quantity of goods supplied and controls the supply of goods to the market. It's another aim would be restricting the supply of goods to competition. Another reason could be greater profit margin.
 - **Forward vertical integration:** It can be described as a movement towards the market outlets. This process is known as forward vertical integration for e.g. large oil companies control most of the world's petrol stations.

Motives of vertical integration:

- Desire to secure a significant number of market outlets and to raise the standards of these outlets.
- Firms are forced to take over some market outlets when a competition steps in this direction.
- Manufacturers are concerned that their products reach the public in form in which they were advertised which forces them for vertical integration.

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- To achieve economies of scale.

Problems:

- A manufacturer moving backwards to acquire the sources of raw materials may find itself with a rapidly depreciating asset if a superior substitute material is available
- During recession the price of raw materials may fall in the world, so the firm integrated with the primary industry would face huge losses.

Horizontal integration: Merger of firms engages in producing the same kind of good or service and at the same stage of production is described as horizontal integration. This is the most common form of merger.

Motives:

- Market domination
- Rationalization of the industry's capacity, particularly if demand is falling
- Horizontal integration also enables the joint capacity of the merged firms to be operated with a greater degree of specialization
- To obtain economies of scale.

Problems:

- It creates monopoly as number of firms are reduced
- There is a possibility of management diseconomies of scale as the size of the firm increases
- Problems arise when managers from different firms belonging to different traditions, races etc. are brought together to form one management team.

Conglomerate integration: These are mergers between firms producing entirely different types of goods and services.

Motives:

- To obtain diversification of output to reduce the risks of trading
- This arises when firms believe there is little scope for any further growth in the market of the existing products.

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Problems:

- It would not achieve economies of scale
- Coordinating a range of products is likely to prove challenging and will require good management.

Multinationals

These are company's which produce in more than one countries. They are sometimes referred as multinational corporations or transnational corporation. Most multinational companies come from industrialized countries.

Advantages:

- Produce and selling in a major market abroad will reduce a firms transport costs.
- Enable it to keep in close contact with that market and may overcome any resistance to buying products made outside the host country.
- Import restriction is also avoided take advantage of regional assistance or to obtain new raw materials they have to pay low wages in other countries.

Effect on their host countries:

- They may provide employment
- They may bring new and advanced methods of technology and a new approach to management.
- They also contribute to the economic growth and increase exports.
- They assist regional policy as they setup their plants in developing areas.
- Consumers in host countries can benefit from increased competition which multinational companies bring.
- They may eliminate domestic producers so that consumers will not receive the advantage of increased competition
- They may exhaust natural resources at a rapid rate and may not follow as high healthy and safety standards as in their own home countries.
- They may engage in transfer pricing i.e. shifting from high tax countries to low tax countries causing high tax countries to lose revenue.
- They have the ability to shift production in the whole world.

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- They may even use the threat of closing down plants in order to win concession e.g. health and safety standards.

Diagrams to be added

1. Separate diagrams for EOS & DEOS
2. Internal vs external
3. AVC & AFC



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